# Financial Statistics Project

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## 1 Garch Family Models of Volatility Modeling for Volvo (Track A)

We use daily data for the VOLVO stock (Track A) covering the period from January 2003 to December 2003. The dataset contains daily closing prices as well as intraday (10-minute) observations, which will be exploited later in the analysis. Figure 1 shows fluctuations and non-stationary behavior of daily closing prices of VOLVO stock, which is typical for financial asset prices. Figure 2 presents daily corresponding log-returns. The return series oscillates around zero and has clear volatility clustering. Therefore we use GARCH family models to capture conditional heteroskedasticity.

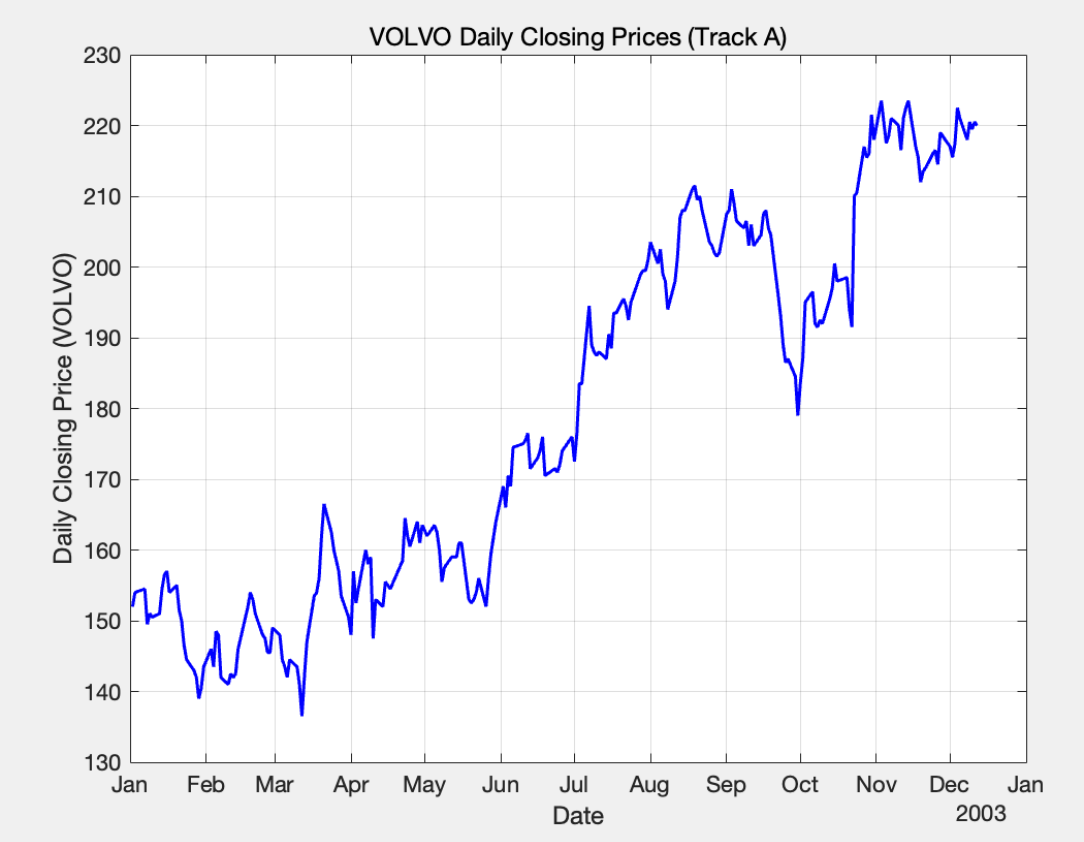


Figure 1: Daily closing prices of VOLVO stock (Track A).

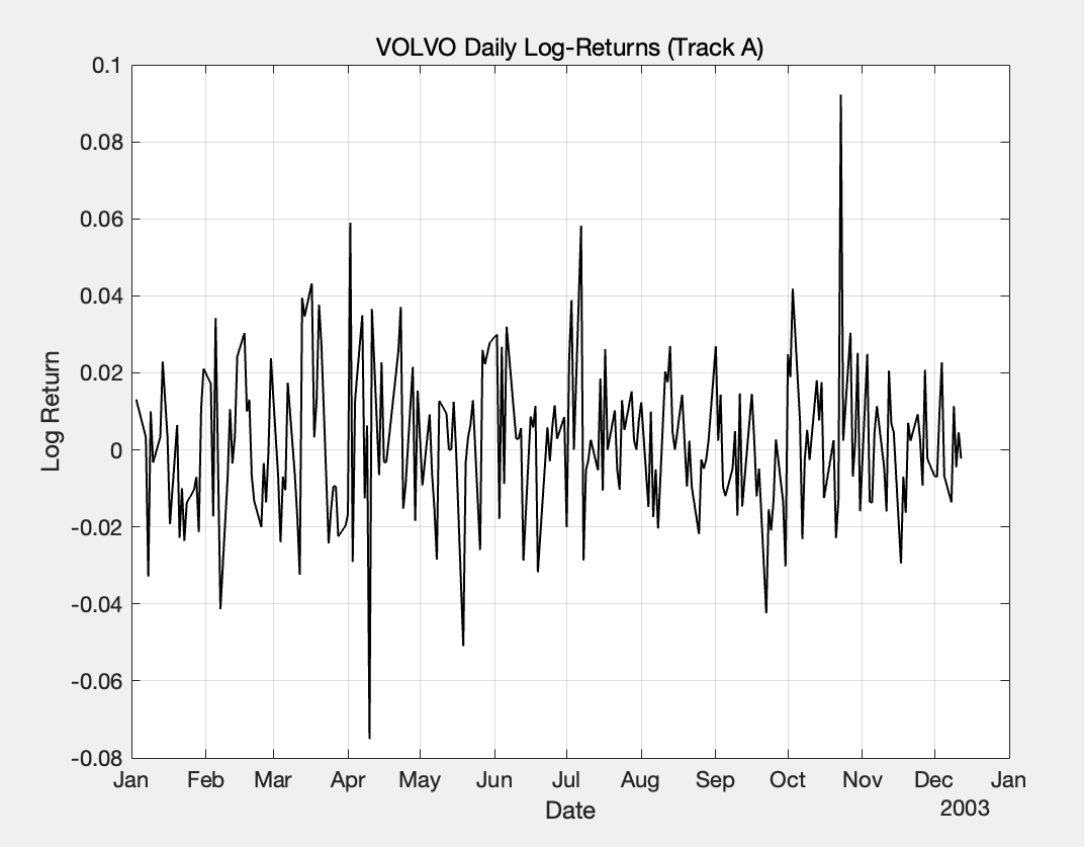


Figure 2: Daily log-returns of VOLVO stock (Track A).

We then first estimate using the EGARCH$(1;1)$ model under both standard Gaussian innovations and heavy-tailed innovations using the `lnl_spec` likelihood provided by Magnus Wiktorsson [2], in order to allow for asymmetric volatility responses and excess kurtosis in returns. However, in both cases the estimation results are unsatisfactory: the EGARCH likelihood fails to evaluate properly and several coefficients exhibit very large standard errors, indicating numerical instability with NaN log-likelihoods. Therefore, we proceed by modeling the returns using a standard GARCH$(1;1)$ model.

We next model daily log-returns using a GARCH$(1;1)$ model under both Gaussian and heavy-tailed innovations. Parameter estimated using the `MLmax` function provided by Magnus Wiktorsson [2]. The return process is given by

$$r_t = \quad + \ ''_t; \qquad ''_t = \ ^{\mathrm{p}}\overline{h_t}\, z_t;$$

with conditional variance

$$h_t = \ ! \ + \quad ''^2_{t\ 1} + \quad h_{t\ 1}:$$

Under Gaussian innovations,

$$z_t \quad N\,(0;1):$$

Under heavy-tailed innovations,

$$z_t \quad \mathrm{lnl\_spec}(c_{\mathrm{Tail}};\, f_{\mathrm{Tail}});$$

allowing for excess kurtosis in returns.

**Gaussian GARCH(1,1).** Under Gaussian innovations, the results are reported in table below with a with a log-likelihood value of $\ln L = 600.10$:

| Parameter | Estimate | SE | $CI_{0.025}$ | $CI_{0.975}$ |
|---|---|---|---|---|
| | 0.0014503 | 0.0013142 | 0.0011256 | 0.0040261 |
| $\omega$ | $3.72 \quad 10^{-5}$ | $4.62 \quad 10^{-5}$ | $5.35 \quad 10^{-5}$ | $1.28 \quad 10^{-4}$ |
| | 0.044748 | 0.043441 | 0.040397 | 0.12989 |
| | 0.85887 | 0.14978 | 0.56531 | 1.1524 |

**Heavy-tailed GARCH(1,1).** Under heavy-tailed innovations using the `lnl_spec` density the results are reported in table below with a with a log-likelihood value of $\ln L = 604.99$:

| Parameter | Estimate | SE | $CI_{0.025}$ | $CI_{0.975}$ |
|---|---|---|---|---|
| | 0.0009671 | 0.0011983 | 0.0013816 | 0.0033157 |
| $\omega$ | $4.06 \quad 10^{-5}$ | $6.41 \quad 10^{-5}$ | $8.51 \quad 10^{-5}$ | $1.66 \quad 10^{-4}$ |
| | 0.051088 | 0.061361 | 0.069179 | 0.17135 |
| | 0.84356 | 0.20660 | 0.43863 | 1.2485 |
| $c_{\text{Tail}}$ | 0.23800 | 0.071557 | 0.097745 | 0.37825 |
| $f_{\text{Tail}}$ | 2.0961 | 3.1354 | 4.0494 | 8.2415 |

**Likelihood–ratio test.** To formally compare the Gaussian and heavy-tailed specifications, we conduct a likelihood–ratio (LR) test defined by

$$\text{LR} = 2\,\ln L_{\text{HT}} \quad \ln L_{\text{Gaussian}}\,.$$

Using the estimated log-likelihood values,

$$\text{LR} = 2(604.99 \quad 600.10) = 9.78.$$

Under the null hypothesis of Gaussian innovations, the LR statistic is asymptotically $^2(2)$ distributed. The resulting $p$-value of 0.0075 below the 5% significance level leads to rejection of the Gaussian GARCH(1,1) model. This means the heavy-tailed innovation model improved the fitting of excess kurtosis of VOLVO log returns.

**Prediction intervals and interval evaluation.** We next evaluate the predictive performance of the heavy-tailed GARCH(1,1) model by constructing the one-step-ahead prediction intervals for daily log-returns. Using the estimated tail parameters from the `lnl_spec` innovation density, the 97.5% quantile of the standardized distribution is estimated numerically as

$$\hat{q}_{0.975} = 2.0554.$$

This yields time varying 95% one-step-ahead prediction intervals of the form

$$_t \quad \hat{q}_{0.975} \sqrt{h_t},$$

Figure 3 illustrates these intervals together with the realized daily log-returns. The majority of observations lie within the bands, with exceedances occurring primarily during periods of extreme market movements.

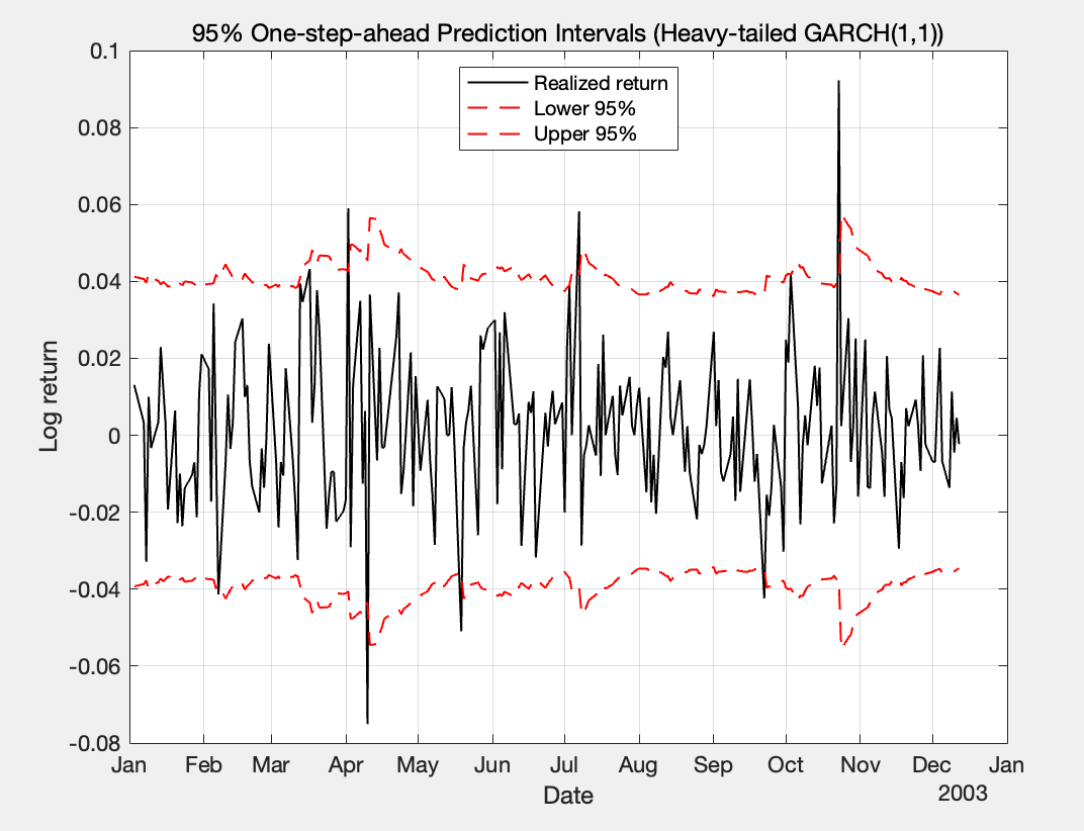


Figure 3: One-step-ahead 95% prediction intervals for VOLVO daily log-returns obtained from the heavy-tailed GARCH(1; 1) model.

For the final in-sample observation, the one-step-ahead forecast interval for the next trading day $r_{T+1}$ is given by

$$\mathrm{PI}_{0:95}(r_{T+1}) = [\ \ 0{:}0342;\ 0{:}0362]$$

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To formally test the interval forecasts, we apply the Christoffersen [3] conditional coverage test, which jointly evaluates correct unconditional coverage and independence of violations. The test statistic is given by

$$\mathrm{LR}_{\mathrm{CC}} = 0{:}5592 \quad (\mathrm{df} = 2);$$

with an associated $p$-value of 0:7561. Consequently, we do not reject the null hypothesis of correct conditional coverage at the 5% significance level. This indicates that the heavy-tailed GARCH(1; 1) model produces well prediction intervals that achieve the nominal coverage probability and do not exhibit clustering of exceedances.

GARCH-X(1,1) with bi-power variation (BPV). We extend the standard GARCH (1; 1) model by incorporating an exogenous volatility regressor constructed from intra-day data, namely bi-power variation (BPV). The model is

$$r_t = \quad + \ ''_t; \qquad ''_t = \sqrt{h_t}\, z_t;$$

$$h_t = \ ! \ + \quad ''^2_{t\ 1} + \quad h_{t\ 1} + \quad \mathrm{BPV}_{t\ 1};$$

where $fz_t g$ are i.i.d. standardized innovations following a heavy-tailed density given by `lnl_spec`. We use the BPV measure that is computed from 10-minute intra-day log-returns $r_{t;j}$ as

$$\mathrm{BPV}_t = \frac{}{2} \sum_{j=2}^{M} r_{t;j} \quad r_{t;j\ 1};$$

with $M$ intra-day intervals per day. Bi-power variation is used instead of daily squared returns because it provides a jump-robust and less sensitive to extreme returns and microstructure intra-day noise. [1] This is particularly suitable for a low-volatility stock such as VOLVO, where large daily returns are rare. Using maximum likelihood estimation with `MLmax` under heavy-tailed innovations, we obtain the fitted GARCH-X model with 95% confidence interval in table below:

| Parameter | Estimate | SE | $\mathrm{CI}_{0:025}$ | $\mathrm{CI}_{0:975}$ |
|---|---|---|---|---|
| | 0:00079281 | 0:0011927 | 0:001545 | 0:0031306 |
| | 0:016232 | 0:064406 | 0:11000 | 0:14247 |
| | 0:51661 | 0:20694 | 0:11101 | 0:92220 |
| | 0:29971 | 0:29998 | 0:28825 | 0:88767 |
| $c_{\mathrm{Tail}}$ | 0:23572 | 0:071582 | 0:095417 | 0:37602 |
| $f_{\mathrm{Tail}}$ | 2:82 | 4:6268 | 6:2485 | 11:889 |

Note that we fix the intercept $! = 0{:}0001$ for numerical stability. and the maximized log-likelihood is $\ln L = 606{:}3414$.

**Prediction intervals from heavy-tailed GARCH-X(1;1) with BPV** We next evaluate the predictive performance of the heavy-tailed GARCH-X(1;1) model. Using the estimated parameters, the 97.5% quantile of the standardized innovation distribution is $\hat{q}_{0:975} = 2{:}0596$, which yields the one-step-ahead 95% prediction interval for the next return

$$\left[ \quad_{T+1} \quad \hat{q}_{0:975} \sqrt{h_{T+1}};\quad_{T+1} + \hat{q}_{0:975} \sqrt{h_{T+1}} \right] = [\ 0{:}0349;\ 0{:}0365]:$$

Figure 4 displays the realized daily log-returns together with the time-varying 95% one-step-ahead prediction intervals. Compared with the standard GARCH(1;1) model, GARCH-X exhibits greater flexibility in its interval dynamics, as the inclusion of BPV allows the conditional variance to respond directly to high-frequency volatility information. This results in prediction bands that adapt more to changes in market conditions rather than reacting only through past squared daily returns.

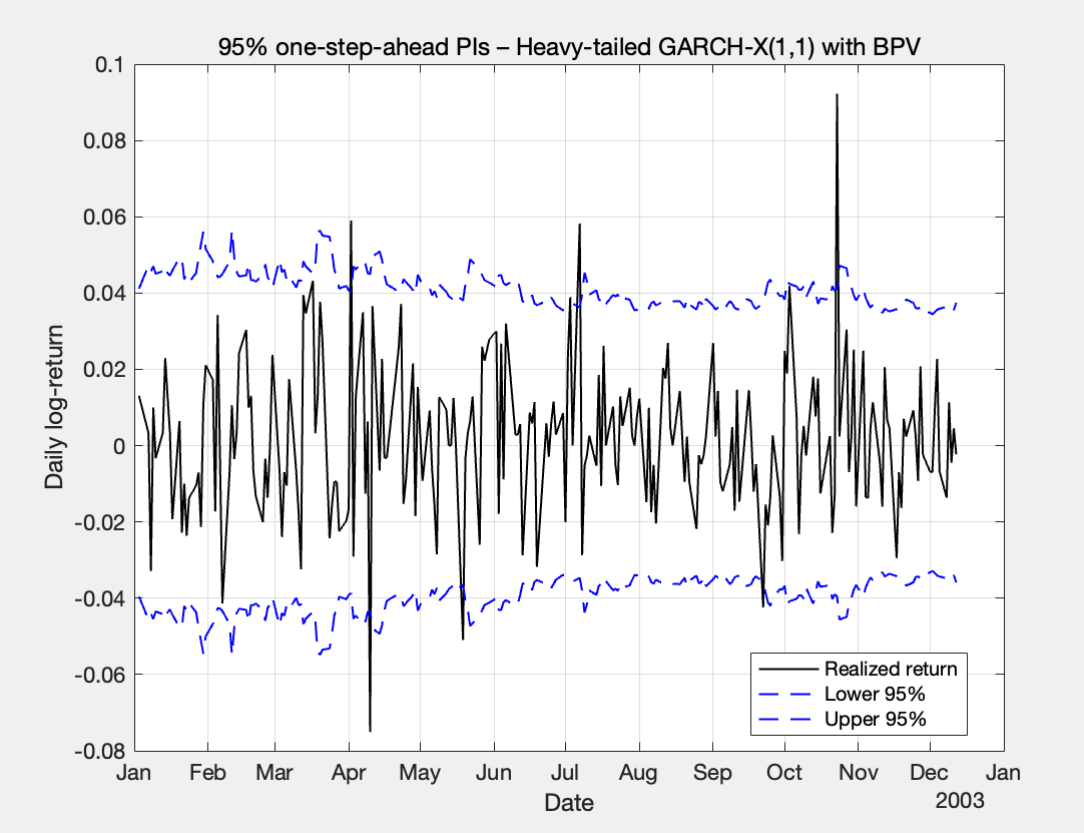


Figure 4: 95% one-step-ahead prediction intervals from the heavy-tailed GARCH-X(1; 1) model with bipower variation (BPV).

Again just like GARCH(1,1) model, we apply the Christoffersen (1998) conditional coverage tests. At the 5% significance level, the unconditional coverage test ($LR_{UC} = 2.48$, $p = 0.116$), the independence test ($LR_{IND} = 0.43$, $p = 0.514$), and the joint conditional coverage test ($LR_{CC} = 2.90$, $p = 0.234$) all fail to reject their respective null hypotheses.

This indicates that the GARCH-X model delivers also a well-calibrated prediction intervals with correct coverage and no evidence of violation clustering.

**LR test: GARCH vs. GARCH–X.** To test whether bipower variation adds explanatory power, we test $H_0$ : = 0 against $H_1$ : 6= 0 using

$$LR = 2 \ln L_{GARCH\text{-}X} \quad \ln L_{GARCH} \qquad {}^2(1) \text{ under } H_0:$$

The test yields $LR = 2.7037$ with $p = 0.1001$, so we do not reject $H_0$ at the 5% significance level. Hence, the inclusion of BPV does not lead to a statistically significant in-sample improvement over the standard GARCH(1,1) model in this dataset.

**Summary** Both the heavy-tailed GARCH(1,1) and GARCH-X(1,1) models produce valid 95% prediction intervals and all passed the Christoffersen conditional coverage tests. However, Figures 3 and 4 show that the GARCH-X model allows for more time-varying dynamics in confidence intervals, indicating greater model flexibility and adaptability relative to the standard GARCH(1,1), despite the absence of statistically significant test improvements in formal interval forecast tests.

# 2 Option Calibration and Evaluation Framework with Different Models (Track A)

In this part we study a calibration and out-of-sample evaluation problem for option pricing models using real market data from OMXS30. The objective is to assess how well different models fit observed option prices and how accurately they can predict prices across strikes, maturities, and time.

**Hold-out sample construction** To enable out-of-sample evaluation, we first reserve a subset of option contracts at each time snapshot as a hold-out sample for later analysis. For each time snapshot $t = 1;:::;T$, we first filter the option panel by removing invalid observations so that we retain only contracts with finite inputs and strictly positive values for MID price, strike, maturity,and finite interest rates. All model calibrations and out-of-sample evaluation are conducted using MID option prices, which provide a balanced proxy between bid and ask quotes and mitigate microstructure noise.

We construct the hold-out sample via stratified sampling across both moneyness and time-to-maturity dimensions to ensure that the evaluation set is representative of the full option panel.

Moneyness is measured by the ratio $K=S$, and contracts are grouped into five buckets using the fixed cutoffs

$$[0;0:90);\ [0:90;0:97);\ [0:97;1:03);\ [1:03;1:10);\ [1:10;\ 1\ ):$$

Time-to-maturity is grouped into three buckets based on empirical quantiles of at each snapshot, using cut points at $(0;0:33;0:66;1)$. Within each non-empty moneyness–maturity bin, we randomly assign 20% of the contracts to the hold-out set for later interpolaiton and extrapolation tests of performance. Note: To ensure comparability across models, the random hold-out assignment is generated once using a fixed random seed and reused in all subsequent model evaluations. Selected contracts per bin is displayed in the following table:

Table 1: Number of hold-out option contracts per moneyness and maturity bin

| Moneyness | Short (Q1) | Medium (Q2) | Long (Q3) |
|---|---|---|---|
| Deep ITM | 87 | 39 | 124 |
| ITM | 114 | 224 | 119 |
| ATM | 108 | 209 | 98 |
| OTM | 112 | 257 | 113 |
| Deep OTM | 108 | 168 | 322 |

Note that the moneyness here is defined using quantiles of the ratio K/S, pooling calls and puts. As a result, for example, the labels "ITM" and "OTM" should be interpreted as relative distance from at-the-money rather than payoff-based moneyness of call/put opintions. Options with the same K/S can therefore include both calls and puts with opposite intrinsic values, but similar distance from ATM.

**State-space estimation via the Unscented Kalman Filter** For all three option pricing models: Black Scholes (BS), Heston, and Bates, we use the Unscented Kalman

Filter (UKF) to estimate time-varying latent states. The UKF is particularly well suited for this task because option prices are nonlinear functions of volatility and structural parameters, and several key quantities such as volatility, stochastic variance, and jump intensities are not directly observable. In contrast to ordinary least squares (OLS), which requires observable regressors, linear measurement equations, and static parameters, the UKF naturally accommodates nonlinear state-space models with latent and dynamically evolving states.[5]. We skip the algorithms description here for briefing, the algorithms can be found in both [5] and [2]. The filtered parameter paths can then be directly passed into the provided `opt_price` routine provided by [2] for interpolation and time-extrapolation evaluation.

**Black Scholes benchmark with time-varying volatility** We begin the option calibration exercise with the BS model, which serves as a natural benchmark. Under the Black–Scholes framework, the underlying asset price follows

$$dS_t = rS_t\,dt + \ _tS_t\,dW_t;$$

where $r$ is the risk-free interest rate and $\ _t$ denotes the (time-varying) instantaneous volatility.

Given this dynamics, the price of a European call/put option at time $t$ with strike $K$ and time to maturity can then be calculated using `opt_price`. Instead of assuming constant volatility, we allow $\ _t$ to evolve over time and estimate it as a latent state using the Unscented Kalman Filter (UKF), based on observed option prices. Note that in the Black Scholes model, the latent volatility is filtered in logarithmic form as $x_t = \log(\ _t)$, ensuring positivity of $\ _t$ throughout the UKF estimation.

Figure 5 shows the filtered BS volatility path. The estimated volatility clearly varies over time and exhibits persistent shifts. This time variation in volatility motivates the use of more flexible stochastic volatility models such as Heston.